

READINESS ASSESSMENT EXAM

Computer Science & Engineering Foundations · 60 Questions · 20 Criteria

Student Name:	_____	Date:	_____
Student ID:	_____	Section:	_____

- This exam contains **60 questions** distributed across **20 criteria**, with 3 questions each (Easy · Medium · Hard).
- **Scoring per criterion:** Easy = 1 pt, Medium = 2 pts, Hard = 3 pts | Max per criterion = **6 pts** | Total max = **120 pts**.
- **Level thresholds per criterion:** 0–2 → Weak | 3–4 → Intermediate | 5–6 → Strong.
- Circle the letter of the single best answer for each question.
- Time allowed: **90 minutes**.

Criterion	Easy (×1)	Medium (×2)	Hard (×3)	Score /6	Level
1. Basic Analysis and Algorithmic Methods	■ Correct	■ Correct	■ Correct	/ 6	_____
2. Algorithms, Design, and Development	■ Correct	■ Correct	■ Correct	/ 6	_____
3. Program Representation, Language, and ...	■ Correct	■ Correct	■ Correct	/ 6	_____
4. Fundamental Programming Concepts	■ Correct	■ Correct	■ Correct	/ 6	_____
5. Basic Type Systems	■ Correct	■ Correct	■ Correct	/ 6	_____
6. Object-Oriented Programming	■ Correct	■ Correct	■ Correct	/ 6	_____
7. Fundamentals of Data Structures	■ Correct	■ Correct	■ Correct	/ 6	_____
8. Database Systems and Data Models	■ Correct	■ Correct	■ Correct	/ 6	_____
9. Software Design, Construction, and Ver...	■ Correct	■ Correct	■ Correct	/ 6	_____
10. Software Processes, Software Managemen...	■ Correct	■ Correct	■ Correct	/ 6	_____
11. Concurrency, Scheduling, Dispatch	■ Correct	■ Correct	■ Correct	/ 6	_____
12. Operating System Principles	■ Correct	■ Correct	■ Correct	/ 6	_____
13. Reliable Data Delivery, Routing, and N...	■ Correct	■ Correct	■ Correct	/ 6	_____

14. Introduction to Networking and Communi...	■ Correct	■ Correct	■ Correct	/ 6	_____ _
15. Graphs, Trees, and Discrete Probabilit...	■ Correct	■ Correct	■ Correct	/ 6	_____ _
16. Proof Techniques and Basics of Computa...	■ Correct	■ Correct	■ Correct	/ 6	_____ _
17. Basic Logic, Sets, Relations, and Func...	■ Correct	■ Correct	■ Correct	/ 6	_____ _
18. Machine Organization on the Basis of A...	■ Correct	■ Correct	■ Correct	/ 6	_____ _
19. Architecture of Memory Systems	■ Correct	■ Correct	■ Correct	/ 6	_____ _
20. Digital Logic, Digital Systems	■ Correct	■ Correct	■ Correct	/ 6	_____ _
TOTAL				/ 120	_____

Criterion 1 — Basic Analysis and Algorithmic Methods

Q1 ● Easy

Weight: 1 pt

Which of the following problems is typically solved using dynamic programming?

- A. Binary Search
- B. Quick Sort
- C. Fibonacci Sequence Calculation with Memoization
- D. Depth-First Search

Answer: ____

Q2 ● Medium

Weight: 2 pts

Which problem demonstrates dynamic programming? Linear Search

- A. Quick Sort
- B. Longest Common Subsequence
- C. BFS

Answer: ____

Q3 ● Hard

Weight: 3 pts

Which standard complexity class describes logarithmic time algorithms? $O(\log n)$

- A. $O(n)$
- B. $O(n^2)$
- C. $O(2^n)$

Answer: ____

Criterion 2 — Algorithms, Design, and Development

Q4 ● Easy

Weight: 1 pt

Which of the following is true about heuristic algorithms?

- A. Guarantee optimal results
- B. Provide fast approximate solutions
- C. Only for sorting
- D. Only for small datasets

Answer: ____

Q5 ● Medium

Weight: 2 pts

Which of the following describes a greedy algorithm's decision process? Recursively explores all possibilities

- A. Selects the best local option at each step
- B. Divides the problem into equal subproblems
- C. Uses random choices to find solutions

Answer: ____

Q6● **Hard**

Weight: 3 pts

Which of the following is true about Big Theta notation?

- A. Gives only an upper bound
- B. Gives only a lower bound
- C. Gives a tight bound on growth rate
- D. Only used in sorting

Answer: ____

Criterion 3 — Program Representation, Language, and Interpretation

Q7● **Easy**

Weight: 1 pt

Which programming concept automatically frees unused memory?

- A. Manual deallocation
- B. Stack pointer adjustment
- C. Garbage collection
- D. Heap resizing

Answer: ____

Q8● **Medium**

Weight: 2 pts

Which section of memory is generally not manually managed by the programmer?

- A. Stack
- B. Heap
- C. Text section and global data (mostly)
- D. None of the above

Answer: ____

Q9● **Hard**

Weight: 3 pts

Syntax errors are usually detected at:

- A. Runtime
- B. Heap allocation
- C. Compile time or parsing
- D. Stack allocation

Answer: ____

Criterion 4 — Fundamental Programming Concepts

Q10 ● Easy

Weight: 1 pt

Garbage collection is most often associated with which type of programming languages?

- A. Assembly only
- B. C with manual memory management
- C. Languages with automatic memory management (e.g., Java, Python)
- D. Languages without runtime

Answer: ____

Q11 ● Medium

Weight: 2 pts

Which section of memory stores global variables in most programming languages?

- A. Stack
- B. Heap
- C. Global data section
- D. Text section

Answer: ____

Q12 ● Hard

Weight: 3 pts

Which tool can generate human-readable documentation from code?

- A. Interpreter
- B. Expression optimizer
- C. Documentation generator
- D. Compiler

Answer: ____

Criterion 5 — Basic Type Systems

Q13 ● Easy

Weight: 1 pt

Which section of memory is used for function call management?

- A. Heap
- B. Global data
- C. Stack
- D. Text section

Answer: ____

Q14 ● Medium

Weight: 2 pts

Which of the following is true about interpreters?

- A. They generate machine code before execution
- B. They execute source code directly line by line
- C. They always optimize expressions
- D. They only generate documentation

Answer: ____

Q15● **Hard**

Weight: 3 pts

Which of the following is true about syntax checking?

- A. It evaluates expressions
- B. It detects memory leaks
- C. It ensures the code conforms to the language grammar
- D. It frees unused memory

Answer: ____

Criterion 6 — Object-Oriented Programming

Q16● **Easy**

Weight: 1 pt

Which of the following is a common task of an interpreter?

- A. Generate AST
- B. Free heap memory
- C. Execute code line by line
- D. Optimize expressions

Answer: ____

Q17● **Medium**

Weight: 2 pts

What is the primary purpose of an interpreter?

- A. To generate documentation
- B. To optimize memory usage
- C. To execute source code directly
- D. To create an AST

Answer: ____

Q18● **Hard**

Weight: 3 pts

Which of the following tools executes code without generating machine code?

- A. Compiler
- B. Interpreter
- C. Expression optimizer
- D. Documentation generator

Answer: ____

Criterion 7 — Fundamentals of Data Structures

Q19 ● Easy

Weight: 1 pt

Which of the following tree data structures is not a balanced binary tree?

- A. Splay tree
- B. B-tree
- C. AVL tree
- D. Red-black tree

Answer: ____

Q20 ● Medium

Weight: 2 pts

Which of the following is the simplest data structure that supports range searching?

- A. AA-trees
- B. K-d trees
- C. Heaps
- D. binary search trees

Answer: ____

Q21 ● Hard

Weight: 3 pts

What is the use of the bin data structure?

- A. to have efficient traversal
- B. to have efficient region query
- C. to have efficient deletion
- D. to have efficient insertion

Answer: ____

Criterion 8 — Database Systems and Data Models

Q22 ● Easy

Weight: 1 pt

A database system typically consists of:

- A. Hardware, Software, Data, and Users
- B. Software, Compiler, and Files
- C. Hardware and Operating System only
- D. Files and File Systems only

Answer: ____

Q23 ● Medium

Weight: 2 pts

A clustering index is defined on the fields which are of type:

- A. Primary key
- B. Foreign key
- C. Non-key, sorting field
- D. Non-key, non-sorting field

Answer: ____

Q24 ● **Hard**

Weight: 3 pts

Which part of the DBMS ensures atomicity, consistency, isolation, and durability?

- A. Transaction Processor
- B. Query Executor
- C. Access Methods
- D. Data Modeler

Answer: ____

Criterion 9 — Software Design, Construction, and Verification

Q25 ● **Easy**

Weight: 1 pt

A primary key in a relation must be:

- A. Unique and not null
- B. Non-unique and null
- C. Optional
- D. Repeated in multiple tables

Answer: ____

Q26 ● **Medium**

Weight: 2 pts

Which of the following is not a component of a database system?

- A. Database
- B. Database Management System (DBMS)
- C. Compiler
- D. Database users

Answer: ____

Q27 ● **Hard**

Weight: 3 pts

Which component stores metadata about the database structure?

- A. Data Dictionary
- B. Data File
- C. Query Processor
- D. Transaction Log

Answer: ____

Criterion 10 — Software Processes, Software Management

Q28 ● Easy

Weight: 1 pt

Which of the following is a declarative query?

- A. SELECT * FROM Students WHERE GPA > 3.0;
- B. FOR each student DO print record;
- C. LOOP through Students;
- D. INSERT using loop iteration

Answer: ____

Q29 ● Medium

Weight: 2 pts

Which function is responsible for restoring a database after a crash?

- A. Backup
- B. Recovery
- C. Logging
- D. Replication

Answer: ____

Q30 ● Hard

Weight: 3 pts

Which of the following is a dense index?

- A. Clustering Index
- B. Primary Index
- C. Secondary Index
- D. All of the above

Answer: ____

Criterion 11 — Concurrency, Scheduling, Dispatch

Q31 ● Easy

Weight: 1 pt

The relationship between scheduling algorithms and application domains is: The same algorithm works optimally for all domains

- A. Scheduling policies must match the performance and timing requirements
- B. of the specific application domain
- C. Scheduling only affects memory usage
- D. Scheduling algorithms are irrelevant for multithreaded applications

Answer: ____

Q32 ● Medium

Weight: 2 pts

In real-time application domains, the scheduling algorithm must ensure:

- A. Maximum CPU utilization without deadlines
- B. Predictable and timely execution of tasks
- C. Only batch processing
- D. First-Come-First-Served execution for all tasks

Answer: ____

Q33 ● Hard

Weight: 3 pts

A task in the "ready" state is:

- A. Waiting for I/O completion
- B. Ready to run but waiting for CPU allocation
- C. Currently executing on the CPU
- D. Terminated

Answer: ____

Criterion 12 — Operating System Principles

Q34 ● Easy

Weight: 1 pt

Which of the following is an example of middleware?

- A. A file stored on disk
- B. Database connection managers
- C. BIOS firmware
- D. CPU microcode

Answer: ____

Q35 ● Medium

Weight: 2 pts

When a new I/O request arrives for a busy device, it is:

- A. Executed immediately
- B. Added to the device driver's I/O queue
- C. Ignored until the device is free
- D. Sent directly to the CPU registers

Answer: ____

Q36 ● Hard

Weight: 3 pts

Middleware enhances portability by:

- A. Allowing applications to bypass the OS
- B. Abstracting heterogeneity across platforms
- C. Requiring hardware-specific coding
- D. Limiting application communication

Answer: ____

Criterion 13 — Reliable Data Delivery, Routing, and Networking

Q37 ● Easy

Weight: 1 pt

Application software relies on the OS for I/O operations because:

- A. It cannot access I/O devices directly
- B. I/O devices only respond to user commands
- C. System software cannot manage I/O
- D. I/O operations require only application-level drivers

Answer: ____

Q38 ● Medium

Weight: 2 pts

The function of process management in an OS is to:

- A. Manage memory allocation only
- B. Control the execution of programs
- C. Organize file paths
- D. Handle user authentication

Answer: ____

Q39 ● Hard

Weight: 3 pts

Which of the following operations must be executed in kernel mode?

- A. Reading a text file in a text editor
- B. Changing a process's memory layout
- C. Printing characters to the screen buffer
- D. Performing simple arithmetic

Answer: ____

Criterion 14 — Introduction to Networking and Communication

Q40 ● Easy

Weight: 1 pt

In operating system architecture, a logical layer is best described as:

- A. A physical hardware component
- B. A conceptual division that separates system functions
- C. A network-routing algorithm
- D. A low-level machine code instruction

Answer: ____

Q41 ● Medium

Weight: 2 pts

Which of the following is a protection and security function of an OS?

- A. Sorting files alphabetically
- B. Preventing unauthorized access
- C. Compressing user files
- D. Improving system speed

Answer: ____

Q42 ● Hard

Weight: 3 pts

When an application attempts to execute a privileged instruction in user mode, the OS:

- A. Executes it normally
- B. Terminates the application or raises an exception
- C. Automatically switches to kernel mode
- D. Converts it to a non-privileged instruction

Answer: ____

Criterion 15 — Graphs, Trees, and Discrete Probability

Q43 ● Easy

Weight: 1 pt

Number of surjective functions from a 2-element set to a 2-element set is: 1

- A. (A) _____
- B. (B) _____
- C. (C) _____
- D. (D) _____

Answer: ____

Q44 ● Medium

Weight: 2 pts

A relation is an equivalence relation if it is: Reflexive and symmetric

- A. Reflexive and transitive
- B. Symmetric and transitive
- C. Reflexive, symmetric and transitive

Answer: ____

Q45 ● Hard

Weight: 3 pts

Number of equivalence relations on $A = \{a,b,c\}$ is:

- A. (A) _____
- B. (B) _____
- C. (C) _____
- D. (D) _____

Answer: ____

Criterion 16 — Proof Techniques and Basics of Computation

Q46 ● Easy

Weight: 1 pt

A graph with no odd cycles must be:

- A. Complete
- B. Bipartite
- C. Eulerian
- D. Hamiltonian

Answer: ____

Q47 ● Medium

Weight: 2 pts

If 7 people sit around a circular table, the number of distinct seating arrangements is:

- A. $6!$
- B. $7!$
- C. $8!$
- D. $5!$

Answer: ____

Q48 ● Hard

Weight: 3 pts

Number of different binary strings of length 8: 128

- A. 256
- B. 512
- C. 1024

Answer: ____

Criterion 17 — Basic Logic, Sets, Relations, and Functions

Q49 ● Easy

Weight: 1 pt

Number of injective functions from a set with 3 elements to a set with 5 elements: 15

- A. 60
- B. 120
- C. 20

Answer: ____

Q50 ● Medium

Weight: 2 pts

Minimum edges required in a connected graph with n vertices: $n-1$

- A. $n+1$
- B. n^2

Answer: ____

Q51 ● Hard

Weight: 3 pts

Two dice are thrown. Probability of getting a sum of 9 is:

- A. $1/6$
- B. $1/9$
- C. $1/8$
- D. $4/36$

Answer: ____

Criterion 18 — Machine Organization on the Basis of Architecture

Q52 ● Easy

Weight: 1 pt

High cache miss rates negatively impact runtime because they: Require additional compiler optimization

- A. Trigger interrupts during each miss
- B. Force the CPU to fetch data from slower memory levels
- C. Consume all register resources

Answer: ____

Q53 ● Medium

Weight: 2 pts

If a cache has a hit rate of 90%, a hit time of 1 ns, and a miss penalty of 10 ns, the average memory access time is approximately: 1 ns

- A. 1.9 ns
- B. 10 ns
- C. 9 ns

Answer: ____

Q54 ● Hard

Weight: 3 pts

One advantage of using interrupts for data transfers over programmed I/O is: CPU utilization increases since it actively waits for data Data transfers occur without requiring the CPU to constantly check device

- A. status
- B. It eliminates the need for memory
- C. The system no longer requires cache memory

Answer: ____

Criterion 19 — Architecture of Memory Systems

Q55

● Easy

Weight: 1 pt

Dynamic ILP techniques such as out-of-order execution are implemented by: Compiler during program translation

- A. Operating system during scheduling
- B. Hardware at runtime
- C. Application programmer manually

Answer: ____

Q56

● Medium

Weight: 2 pts

Which component of the von Neumann machine is responsible for fetching instructions from memory?
Arithmetic Logic Unit (ALU)

- A. Control Unit (CU)
- B. Data Bus
- C. Input/Output Interface

Answer: ____

Q57

● Hard

Weight: 3 pts

At the assembly level, instructions are expressed using:

- A. Natural language descriptions
- B. Binary numbers only
- C. Symbolic mnemonics and operands
- D. Microcode tables

Answer: ____

Criterion 20 — Digital Logic, Digital Systems

Q58

● Easy

Weight: 1 pt

The use of addressing modes in machine instructions allows the CPU to: Increase clock frequency

- A. Determine how to interpret operand references
- B. Eliminate the need for registers
- C. Simplify the compiler's lexical analysis

Answer: ____

Q59

● Medium

Weight: 2 pts

A typical R-type instruction in a RISC architecture includes fields for: Opcode, source registers, destination register, function code

- A. Only opcode and immediate value
- B. Opcode and memory address
- C. Condition codes and stack pointer

Answer: ____

Q60

● **Hard**

Weight: 3 pts

Instruction-level parallelism (ILP) refers to the ability of a processor to: Execute multiple programs at the same time

- A. Execute multiple instructions from the same program
- B. simultaneously
- C. Increase cache size to hold more instructions
- D. Perform I/O operations in parallel with computation

Answer: ____

ANSWER KEY & SCORING GUIDE

#	Criterion	Diff	Wt	Correct?
1	1. Basic Analysis and Algorithmic Methods	Easy	1	■ Yes ■ No
2	1. Basic Analysis and Algorithmic Methods	Medium	2	■ Yes ■ No
3	1. Basic Analysis and Algorithmic Methods	Hard	3	■ Yes ■ No
4	2. Algorithms, Design, and Development	Easy	1	■ Yes ■ No
5	2. Algorithms, Design, and Development	Medium	2	■ Yes ■ No
6	2. Algorithms, Design, and Development	Hard	3	■ Yes ■ No
7	3. Program Representation, Language, and Inte...	Easy	1	■ Yes ■ No
8	3. Program Representation, Language, and Inte...	Medium	2	■ Yes ■ No
9	3. Program Representation, Language, and Inte...	Hard	3	■ Yes ■ No
10	4. Fundamental Programming Concepts	Easy	1	■ Yes ■ No
11	4. Fundamental Programming Concepts	Medium	2	■ Yes ■ No
12	4. Fundamental Programming Concepts	Hard	3	■ Yes ■ No
13	5. Basic Type Systems	Easy	1	■ Yes ■ No
14	5. Basic Type Systems	Medium	2	■ Yes ■ No
15	5. Basic Type Systems	Hard	3	■ Yes ■ No
16	6. Object-Oriented Programming	Easy	1	■ Yes ■ No
17	6. Object-Oriented Programming	Medium	2	■ Yes ■ No
18	6. Object-Oriented Programming	Hard	3	■ Yes ■ No
19	7. Fundamentals of Data Structures	Easy	1	■ Yes ■ No
20	7. Fundamentals of Data Structures	Medium	2	■ Yes ■ No
21	7. Fundamentals of Data Structures	Hard	3	■ Yes ■ No
22	8. Database Systems and Data Models	Easy	1	■ Yes ■ No
23	8. Database Systems and Data Models	Medium	2	■ Yes ■ No
24	8. Database Systems and Data Models	Hard	3	■ Yes ■ No
25	9. Software Design, Construction, and Verific...	Easy	1	■ Yes ■ No
26	9. Software Design, Construction, and Verific...	Medium	2	■ Yes ■ No
27	9. Software Design, Construction, and Verific...	Hard	3	■ Yes ■ No
28	10. Software Processes, Software Management	Easy	1	■ Yes ■ No
29	10. Software Processes, Software Management	Medium	2	■ Yes ■ No
30	10. Software Processes, Software Management	Hard	3	■ Yes ■ No
31	11. Concurrency, Scheduling, Dispatch	Easy	1	■ Yes ■ No
32	11. Concurrency, Scheduling, Dispatch	Medium	2	■ Yes ■ No
33	11. Concurrency, Scheduling, Dispatch	Hard	3	■ Yes ■ No

34	12. Operating System Principles	Easy	1	■ Yes ■ No
35	12. Operating System Principles	Medium	2	■ Yes ■ No
36	12. Operating System Principles	Hard	3	■ Yes ■ No
37	13. Reliable Data Delivery, Routing, and Netwo...	Easy	1	■ Yes ■ No
38	13. Reliable Data Delivery, Routing, and Netwo...	Medium	2	■ Yes ■ No
39	13. Reliable Data Delivery, Routing, and Netwo...	Hard	3	■ Yes ■ No
40	14. Introduction to Networking and Communicati...	Easy	1	■ Yes ■ No
41	14. Introduction to Networking and Communicati...	Medium	2	■ Yes ■ No
42	14. Introduction to Networking and Communicati...	Hard	3	■ Yes ■ No
43	15. Graphs, Trees, and Discrete Probability	Easy	1	■ Yes ■ No
44	15. Graphs, Trees, and Discrete Probability	Medium	2	■ Yes ■ No
45	15. Graphs, Trees, and Discrete Probability	Hard	3	■ Yes ■ No
46	16. Proof Techniques and Basics of Computation	Easy	1	■ Yes ■ No
47	16. Proof Techniques and Basics of Computation	Medium	2	■ Yes ■ No
48	16. Proof Techniques and Basics of Computation	Hard	3	■ Yes ■ No
49	17. Basic Logic, Sets, Relations, and Function...	Easy	1	■ Yes ■ No
50	17. Basic Logic, Sets, Relations, and Function...	Medium	2	■ Yes ■ No
51	17. Basic Logic, Sets, Relations, and Function...	Hard	3	■ Yes ■ No
52	18. Machine Organization on the Basis of Archi...	Easy	1	■ Yes ■ No
53	18. Machine Organization on the Basis of Archi...	Medium	2	■ Yes ■ No
54	18. Machine Organization on the Basis of Archi...	Hard	3	■ Yes ■ No
55	19. Architecture of Memory Systems	Easy	1	■ Yes ■ No
56	19. Architecture of Memory Systems	Medium	2	■ Yes ■ No
57	19. Architecture of Memory Systems	Hard	3	■ Yes ■ No
58	20. Digital Logic, Digital Systems	Easy	1	■ Yes ■ No
59	20. Digital Logic, Digital Systems	Medium	2	■ Yes ■ No
60	20. Digital Logic, Digital Systems	Hard	3	■ Yes ■ No

Criterion Score → Level		Overall Readiness Score	
Score	Level	Total Score	Overall Level
0 – 2	Weak	0 – 40	Weak
3 – 4	Intermediate	41 – 80	Intermediate
5 – 6	Strong	81 – 120	Strong